1. Exam Pattern & Evaluation Blueprint (70 Marks)

Every question in the Oct/Nov 2025 final exam was selected directly from your Assignments (1–5) and Mid-1 Question Bank. Following your faculty's official evaluation key, here is how we will structure your answers to secure full marks:

Section	Pattern	Sir's Marking Breakdown	How We Will Prepare Each Answer
Part A (10M)	5×2M (1 per Unit, no choices)	1M + 1M: Definition (1M) + 2 Examples/Factors (1M), or a 2-point comparison table (2M).	Crisp 2-line definitions, acronym expansions, and 2-point tables.
Part B (60M) — Split Question	5×12M (Q2 to Q6, 1 per Unit with internal OR choice). Often split into (i) 6M + (ii) 6M.	3M + 3M: Neat labeled diagram or comparison table (3M) + 4–5 key bullet points or example (3M).	ASCII/textbook diagrams you can sketch in 60 seconds + high-scoring keyword bullets.
Part B (60M) — Full Question	Single 12M question on a core architecture or algorithm.	2–4M (Diagram) + 4–5M (Step-by-step working/layers) + 4–5M (Real-world trace or numerical example).	Complete 3-part answers (Diagram → Mechanism → Worked Example).

2. Master Unit-by-Unit Preparation Checklist

We will cover all important syllabus concepts, prioritizing the fresh Assignment/Mid-1 questions (prime rotation candidates for tomorrow) alongside the 2025 exam questions (in case of repeats):

Unit I: Introduction to Networks & Physical Layer (CO1)

- **Reference Models:** OSI 7-Layer Model (layer functions, design issues, and peer-to-peer data transmission/encapsulation), TCP/IP Model layers & protocols, and OSI vs. TCP/IP comparison.
- **Network Types & Topologies:** LAN, MAN, WAN applications & LAN vs. WAN comparison; Topologies (Star, Bus, Ring, Mesh, Tree, Hybrid) with diagrams and limitations.
- **Transmission Media:** Guided Media (Twisted Pair UTP/STP, Coaxial Cable 4-layer structure, Fiber Optic construction & total internal reflection) vs. Unguided Media (Radio waves, Microwaves, Infrared).
- **Multiplexing & Telephone Systems:** FDM, TDM (Synchronous & Statistical), WDM, and CDM with sketches; PSTN architecture/call switching and Mobile Telephone System (GSM / 1G, 2G, 3G).

Unit II: Data Link Layer & MAC Sublayer (CO2)

- **DLL Design & Framing:** Design issues, Services to Network layer, and Framing techniques (Character count, Byte stuffing, Bit stuffing, Coding violations).
- **Error Detection & Correction (Theory + Numericals):** Even/Odd Parity check, 1's Complement Checksum calculation, CRC (polynomial modulo-2 division), and Hamming Code / Minimum Hamming Distance (2t+1 and e+1).
- **Flow Control & ARQ:** Stop-and-Wait protocol and Sliding Window protocols (Go-Back-N vs. Selective Repeat).
- **MAC Protocols & LAN Standards:** Pure vs. Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA (Collision Prevention), Collision-free protocols; IEEE 802.3 Ethernet (MAC frame format, Standard/Fast/Gigabit Ethernet), Token Bus (802.4) vs. Token Ring (802.5), IEEE 802.11/802.16, Bluetooth, Bridges, and VLANs.

Unit III: Network Layer (CO3)

- **Routing Algorithms:** Design issues, Virtual Circuit vs. Datagram subnets, Shortest Path (Dijkstra), Flooding, Distance Vector Routing (with graph table calculation), Link State Routing, Hierarchical Routing, Broadcast routing, OSPF, and BGP.
- **Congestion Control:** General principles, Open Loop vs. Closed Loop control, Choke packets, Load shedding, Jitter control, and Leaky Bucket vs. Token Bucket algorithms.
- **Internetworking & IP:** Tunneling, Fragmentation, IPv4 Header format, IPv4 vs. IPv6 comparison, Classful vs. Classless (CIDR) addressing, Subnetting numericals (calculating subnets, hosts per subnet, and broadcast IPs for 192.168.100.0/26), and ICMP/ARP.

Unit IV: Transport Layer (CO4)

- **Transport Services & Elements:** Addressing (Ports/TSAPs), Multiplexing, Flow control & Buffering, and Crash recovery.
- **UDP, RTP & RPC:** UDP header format, when UDP is preferred over TCP, Real-Time Transport Protocol (RTP), and Remote Procedure Call (RPC).
- **TCP Protocol & Congestion Control:** TCP Segment Header, TCP vs. UDP header comparison, Connection Establishment (3-Way Handshake) & Release, FSM Connection Management, Sliding Window & Silly Window Syndrome, Adaptive Retransmission, TCP Congestion Control (Slow Start, Congestion Avoidance, Fast Retransmit), and DTN/Bundle Protocol.

Unit V: Application Layer (CO5)

- **WWW & HTTP:** WWW Architecture (Client/Server), Hyperlink click-to-page display steps, URL components vs. URI, HTTP Operational Model, Persistent vs. Non-Persistent HTTP, HTTP Request/Response message formats, and HTTP Methods (GET, HEAD, POST, PUT, DELETE, TRACE).
- **DNS:** DNS Namespace & Hierarchy, Domain-to-IP resolution (Recursive vs. Iterative), Name Servers, and DNS Resource Records (5-tuple format).
- **Electronic Mail & Streaming:** Email Architecture (UA, MTA, Final Delivery), RFC 5322 Header fields, SMTP working, MIME (multimedia justification), POP3 vs. IMAP4 comparison, and Streaming Audio/Video.